XAI-CropInsure

Explainable AI for Transparent Crop-Insurance Claim Assessment

DECISION SUPPORT NOTICE: *This report provides machine-learning decision support. It is not an official insurance approval, rejection, or guaranteed payout determination. Final decisions must follow the applicable insurance policy, verified local evidence, and authorized review procedures.*

Claim Reference: CLM-2026-0013	**Report Date:** 2026-10-27 23:42:49
Model: Baseline Random Forest (Historical Outcome) (v1.d93928)	**Workflow Status:** Human review completed

1. Farmer & Agricultural Overview

Farmer Name: Test Farmer Harish	**Policy Ref:** POL-TEST-2026-001
Location: Solapur, Maharashtra	**Insurance Scheme:** Pradhan Mantri Fasal Bima Yojana (PMFBY)
Crop & Season: Cotton (Kharif)	**Cultivated Area:** 3.5 Hectares
Sowing Date: 2025-06-15	**Claim Window:** 2025-07-01 to 2025-11-15

2. Environmental & Remote Sensing Observations

Variable	Observed Value	Reference Baseline	Derived Anomaly / Deficit
Rainfall (mm)	310.0 mm	850.0 mm	540.0 mm deficit
Vegetation Index (NDVI)	0.380	0.680	-0.300 change
Temperature (°C)	35.5 °C	31.0 °C	+4.5 °C anomaly
Soil Moisture (%)	22.0%	Seasonal Benchmark	Field moisture observation

3. Model-Based Assessment & SHAP Explanations

Predicted Pattern Match: Approved (Confidence / Probability: 97.0%)

Feature	Observed	SHAP Impact	Direction	Context
NDVI Vegetation Change	-0.30000000000000004	+0.1174	Positive	Observed value (-0.30000000000000004) increased the model's ...
Relative Rainfall Deficit (%)	0.6352941176470588	+0.0991	Positive	Observed value (0.6352941176470588) increased the model's li...
Temperature Anomaly (°C)	4.5	+0.0755	Positive	Observed value (4.5) increased the model's likelihood of mat...
Soil Moisture (%)	22.0	+0.0510	Positive	Observed value (22.0) increased the model's likelihood of ma...
Rainfall Deficit (mm)	540.0	+0.0271	Positive	Observed value (540.0) increased the model's likelihood of m...
State (Maharashtra)	Maharashtra	-0.0184	Negative	Observed value (Maharashtra) decreased the model's likelihoo...

4. Farmer-Friendly Summary

The machine-learning model indicates that this claim shares characteristics with historically approved claims (estimated pattern match probability: 97.0%).

Factors supporting higher claim likelihood: NDVI Vegetation Change (observed: -0.30000000000000004), Relative Rainfall Deficit (%) (observed: 0.6352941176470588), Temperature Anomaly (°C) (observed: 4.5). These conditions align with patterns of agricultural stress observed in past records.

Factors lowering claim likelihood: State (Maharashtra) (observed: Maharashtra), Cultivated Area (Hectares) (observed: 3.5), Crop Type (Cotton) (observed: Cotton). These indicators suggest adequate moisture, stable vegetation vigor, or mitigating circumstances during the observation window.

Data quality review: All core meteorological and satellite vegetation indicators were present and within normal physical bounds.

IMPORTANT DECISION SUPPORT NOTICE: This automated assessment reflects statistical pattern matching and does NOT constitute a final legal or financial determination. Under PMFBY and agricultural insurance guidelines, an authorized human claim assessor must independently verify local field conditions, loss assessment reports (CCEs), and policy provisions before making any final decision.

5. Authorized Human Review & Assessor Sign-off

Assessor Name: _____	**Official Decision:** [ ] Approved [ ] Rejected [ ] Evid. Req.
Assessor Signature: _____	**Date:** _____
Assessor Notes: _____ _____	